QME References · 01

# BikePulse Gear

Quote and invoice built on the bike, in 3D

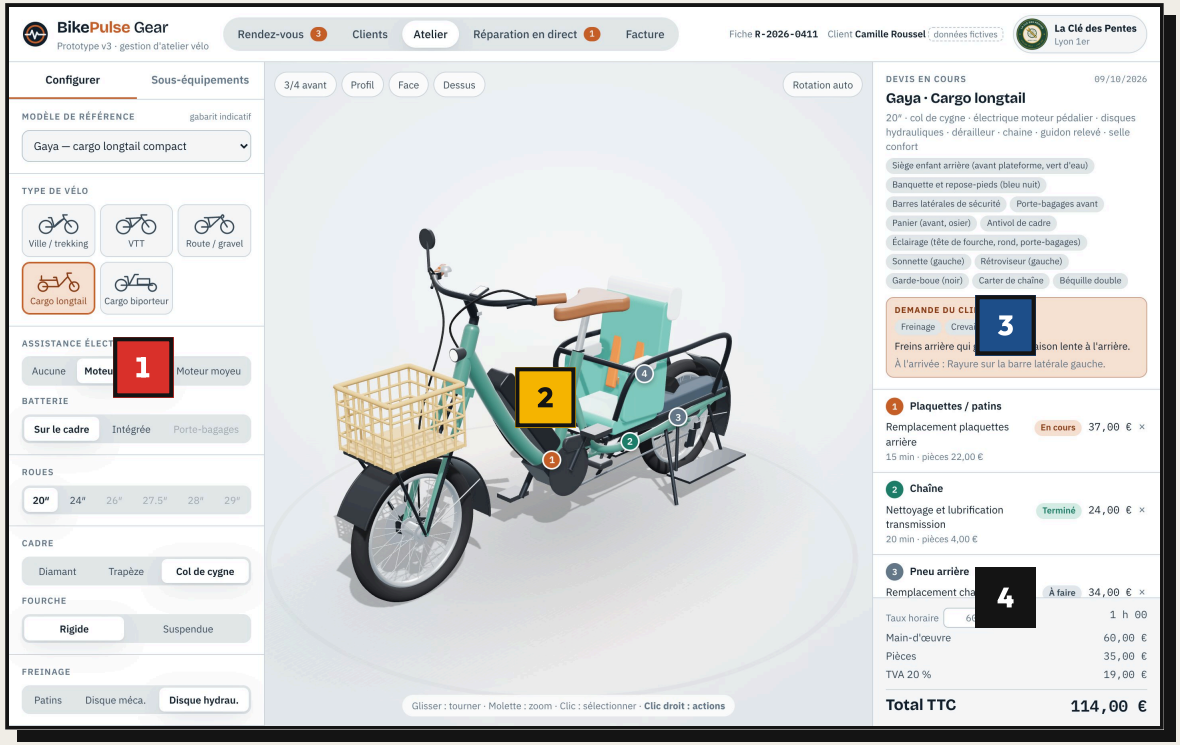
BIKE SHOPS · E-BIKES · CARGO

## 1 The customer's bike

Type, motor, battery, wheels, frame, fork, brakes: each choice rebuilds the 3D model. The reference model serves as a template.

## 2 Click on the part

Click on the bike, right-click for the action wheel: the job is added and a numbered marker pins it to the part.




Workshop screen. Fictitious workshop, customer and prices.

## 3 Customer request

Reasons for the visit and condition on arrival, entered at booking, stay in front of the mechanic.

## 4 A live quote

Time, parts and status for each job (to do, in progress, done). Labor, VAT and total are recalculated live; the invoice follows from it.

### WHAT I DESIGNED

The full workshop flow: booking, bike drop-off, workshop, live repair, invoice. The 3D engine for the bike and its equipment. The presentation website with a narrated video and a form.

### TOOLS

Single-page web app, Three.js for 3D, Netlify website, video edited by script with a synthetic voice-over.

# Bricabulle

Nursery management with accurate billing

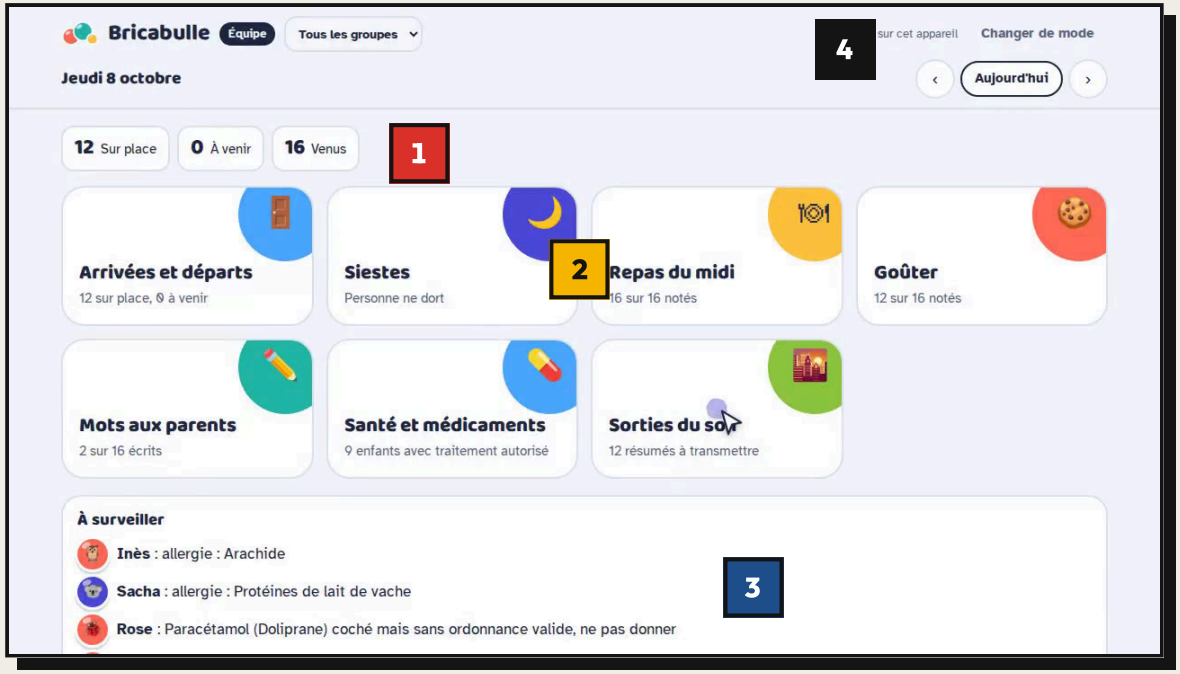
NURSERIES · EARLY CHILDHOOD

## 1 The room at a glance

Children present, expected and already gone: the team always knows who is there.

## 2 One tile per action

Arrivals and departures, naps, meals, snacks, notes to parents: each moment of the day is logged in one tap on the room tablet.




Team tablet. Fictitious children and nursery.

## 3 Health first

Allergies and treatments surface on their own. A medicine ticked without a valid prescription is flagged before it is given.

## 4 Four modes, one tool

Team tablet, parent kiosk at the entrance, online parent portal, management dashboard.

### WHAT I DESIGNED

The link between the schedule booked by parents, the hours actually attended and the invoice. The evening summary for each child. Management oversight: real-time staff ratios, missed check-ins, disputes, schedules and leave.

### TOOLS

Single-page web app, script-generated demo, subtitled video, website and form on Netlify.

# Nexora

Tasks, projects and meetings, with AI agents plugged in

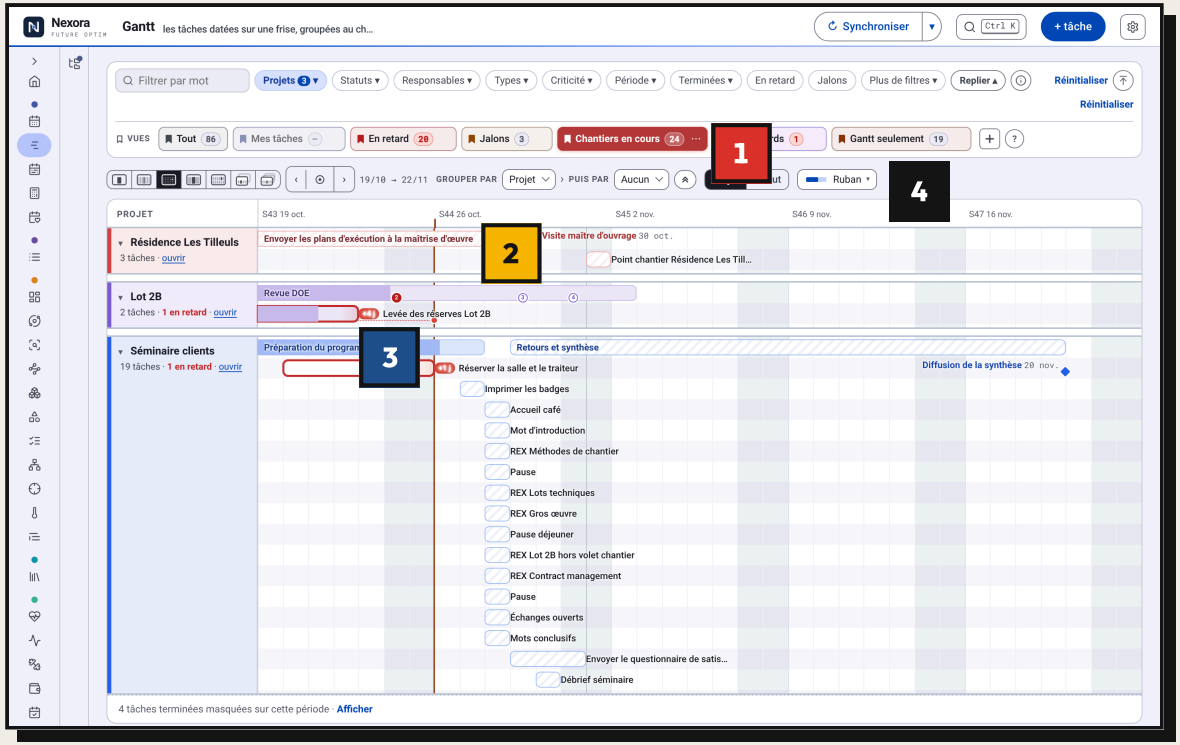
PROJECT MANAGEMENT

## 1 Saved views

A view keeps its filters. One click shows ongoing sites, another shows delays or milestones.

## 2 Milestones and visits

Visits, meetings and deadlines sit on the timeline, within their project. The vertical line marks today.




Gantt view of the app. Fictitious projects, tasks and people.

## 3 Visible delays

An overdue task is circled in red, with its delay in days. The bar shows progress.

## 4 Group, zoom

By project or by status, from week to year. The same data is also available as a list, an orbit or a daily dial.

### WHAT I DESIGNED

A complete app: configurable task types, overlap-free scheduling, meetings and minutes, Google Calendar sync, a connector that lets AI assistants create tasks from incoming emails.

### TOOLS

React, Firebase, Netlify, MCP connector, automated tests in a real browser.

# Drawing approvals

From drawing to approval for construction

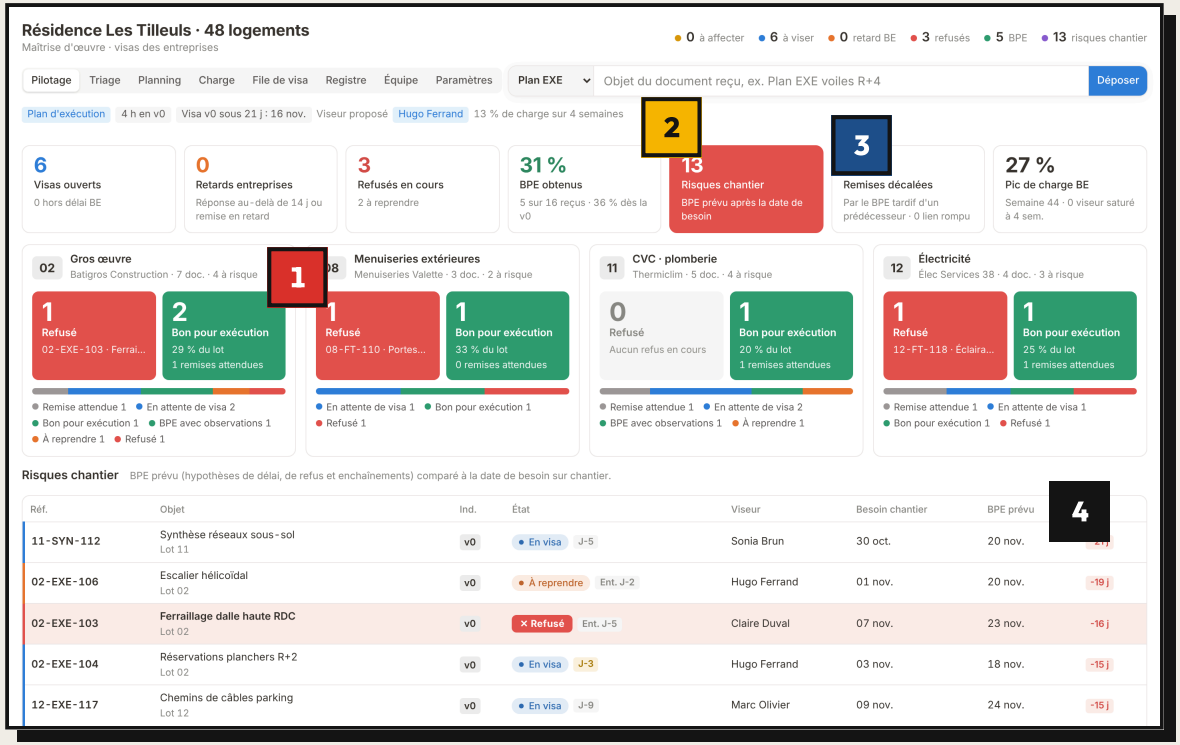
ENGINEERING FIRM · CONFIDENTIAL

## 1 Packages at a glance

Open rejections, approvals for construction and expected submissions, package by package and contractor by contractor.

## 2 Upload and assign

A received document is logged in one line. A reviewer is suggested based on discipline and workload over four weeks.




Overview screen. Fictitious project, contractors and people.

## 3 Site risks

The planned approval date is compared with the date the site needs it. What will hold up the works shows up early.

## 4 The gap in days

Each document shows its revision, reviewer, status and the gap between planned approval and site need.

### WHAT I DESIGNED

An engineering firm's drawing approval register: document upload and assignment, review queue per reviewer, revision planning with rejection probability, workload curve, dependencies between documents, alert when an approval risks arriving after the site needs it.

### TOOLS

Single-page web app, real-time shared database, approval lead-time forecasting.

# Box ABCD

The competition planned in 3D, from floor plan to leaderboard

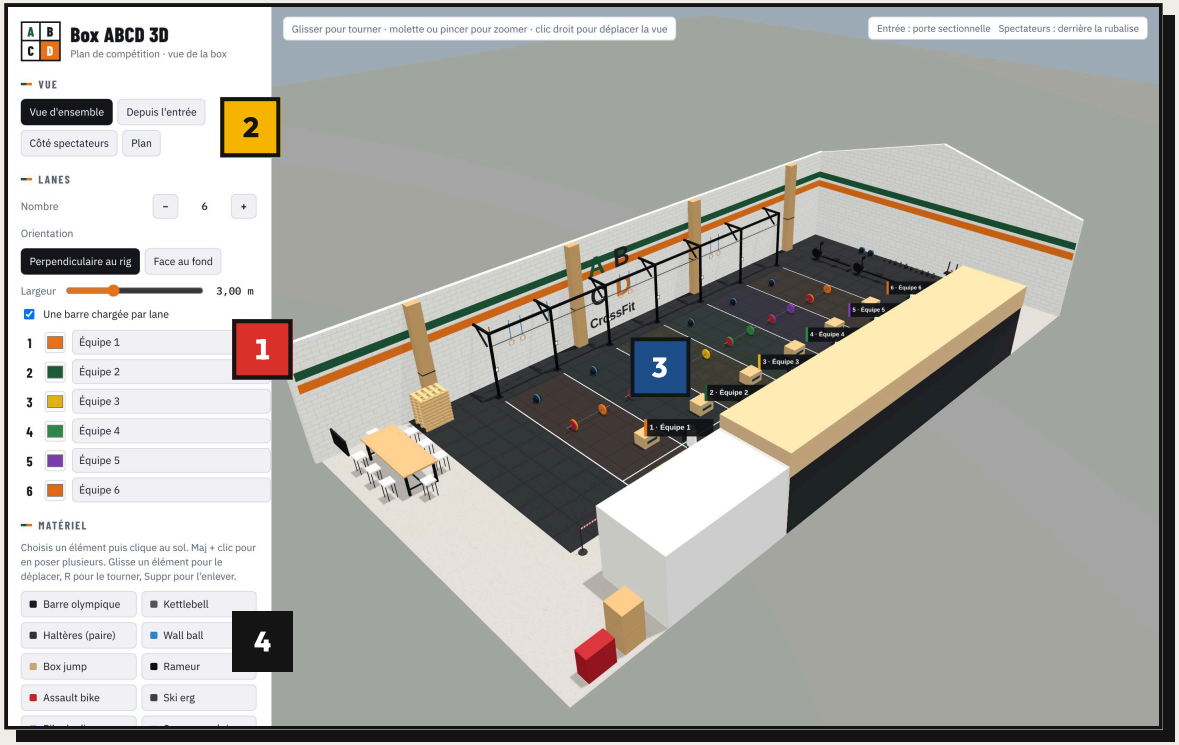
CROSSFIT GYM · COMPETITIONS

### 1 Lanes and teams

Number of lanes, width, orientation and each team's color: the floor redraws with every setting.

### 2 Viewpoints

Overview, from the entrance, from the spectator side or top-down: enough to brief judges and volunteers.




Floor setup view. Anonymized box and teams.

### 3 The box to scale

Rig, loaded barbells, box jumps and spectator area placed to scale, to settle the layout before the big day.

### 4 Equipment to place

Pick an item and click on the floor. Equipment planned, set up and checked is tracked heat by heat.

**WHAT I DESIGNED**

A competition tool for a CrossFit box: 3D floor to scale, lanes and teams, equipment to place and check, projected heats and timer, live leaderboard, TV mode for spectators.

**TOOLS**

Web app, Three.js for 3D, real-time shared database.